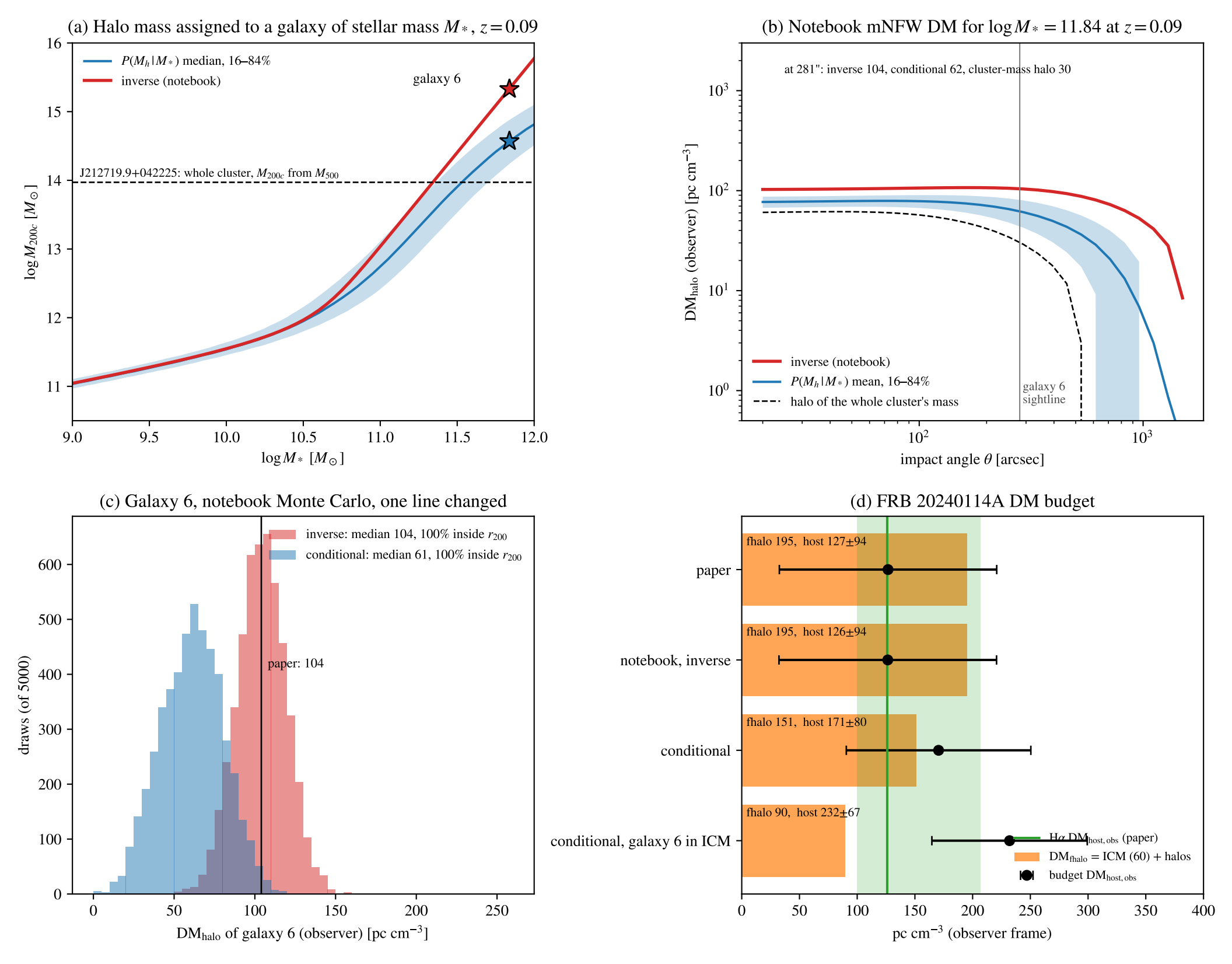
(a) Halo mass assigned to a galaxy of stellar mass M_*, $z = 0.09$
$P(M_h|M_*)$ median, 16–84%
inverse (notebook)
galaxy 6
J212719.9+042225: whole cluster, M_{200c} from M_{500}
$\log M_{200c}$ $[M_\odot]$
$\log M_*$ $[M_\odot]$
(b) Notebook mNFW DM for $\log M_* = 11.84$ at $z = 0.09$
at 281": inverse 104, conditional 62, cluster-mass halo 30
inverse (notebook)
$P(M_h|M_*)$ mean, 16–84%
halo of the whole cluster's mass
galaxy 6 sightline
$\mathrm{DM_{halo}}$ (observer) [pc cm^{-3}]
impact angle θ [arcsec]
(c) Galaxy 6, notebook Monte Carlo, one line changed
inverse: median 104, 100% inside r_{200}
conditional: median 61, 100% inside r_{200}
paper: 104
draws (of 5000)
$\mathrm{DM_{halo}}$ of galaxy 6 (observer) [pc cm^{-3}]
(d) FRB 20240114A DM budget
paper
notebook, inverse
conditional
conditional, galaxy 6 in ICM
fhalo 195, host 127±94
fhalo 195, host 126±94
fhalo 151, host 171±80
fhalo 90, host 232±67
Hα $\mathrm{DM_{host,obs}}$ (paper)
$\mathrm{DM_{fhalo}}$ = ICM (60) + halos
budget $\mathrm{DM_{host,obs}}$
pc cm^{-3} (observer frame)